MILITARY SPECIFICATION

MINE, ANTIPERSONNEL M86, DEMONSTRATION MINE

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembling and packing for the Molded Assembly used in the M86.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-45743	-Soldering, Hand Type, High Reliability Electrical, Electronic, Instrument, Communications, and Radar for Aerospace, and Control Systems, procedures For.
MIL-S-46844	-Solder Bath Soldering of Printed-wiring Assemblies, Automatic Machine Type.
MIL-A-48078	-Ammunition, General Quality Assurance Provisions, General Specification For.
MIL-M-48245	-Molding Compound, Housing and Plug.
MIL-I-48331	-Integrated Circuit, Digital, CMOS Time Base and Special Purpose Function Generator, Monolithic, Silicon.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commander, U.S Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1345
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MIL-B-70731 -Battery, Primary Reserve Ammunition, 8 Milliampere Hour Capacity for Pursuit Deterrent Munition - M86
MIL-R-63993 -Resistor, Thick Film, Fixed and Trimmable.
MIL-A-70780 -Arming Assembly for Pursuit Deterrent Munition, M86
MIL-P-55110 -Printed Wiring Boards, General Spec. For.

STANDARDS

FEDERAL

FED-STD-595 -Color (Requirements for individual color chips).

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 -Test Methods for Electronic and Electrical Component Parts.
MIL-STD-275 -Printed Wiring for Electronic Equipment.
MIL-STD-750 -Test Methods for Semiconductor Devices.
MIL-STD-883 -Test Methods and Procedures for Microelectronics
MIL-STD-1169 -Packaging, Packing and Marketing for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (See 6.7)

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9366698 -Demonstration Mine
9366710 -Packing and Marking for Box, Wirebound for Mine APERS (PDM), M86 or Classroom Model
12910123 -Demonstration Mine with Bandoleer for M86 APERS Mine

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D-695-(85) -Compressive Properties of Rigid Plastics

(Application for copies should be adressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications and standards.

3.2 Molded assembly. The Molded Assembly shall comply with all requirements specified on Dwg. 9366705 and 9366661-2, all associated drawings, and with all requirements in applicable specifications and standards including MIL-M-70780.

3.3 S&A Assembly. (Dwg. 9366668).

3.3.1 Presence of a detonator only in S&A assembly. The S&A assembly shall have the detonator only present and properly positioned and assembled.

3.3.2 Qualification test of electric detonator. The electric detonator serial lot, in the molded assembly lot (see 4.4.1) shall perform satisfactorily as evidenced by high order functioning.

3.4 Function of sensor assembly. (Dwg. 9298569).

3.4.1 Sensor release. The Sensor Assembly when subjected to a pressure of 250 + 25 psig for 1 second maximum while at a temperature of 75°F ° 10°F shall release the ejection spring, sleeve, cap and bobbin with the thread remaining unbroken and attached to the bobbin and post with the breakwire maintaining continuity when tested as specified in 4.5.4.1.

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3.4.2 Tripline deployment. When the Sensor Assembly functions in 3.4.1, the tripline shall deploy a minimum of 16 feet when tested in accordance with 4.5.4.2.

3.4.3 Breakwire function. When the pressure is released and the line pulled, the breakwire shall break at a force of 140 grams, minimum, and 500 grams, maximum, when tested in accordance with 4.5.4.3.

3.4.4 Post retention and thread breaking strength. Following the breakwire function, the post must be retained within the sensor case and the thread must not break when a 540 ram minimum dislodging force is applied in accordance with 4.5.4.4.

3.4.5 Molding material intrusion test. After molding into a test brick, the sensor assembly when subjected to 250 + 25 psig for 1 second maximum and while at a temperature of 75°F ± 10°F shall release the cap, sleeve and bobbin assembly when tested as specified in 4.5.4.6.

3.5 Functioning of electronic circuitry. After molding into the Molded Assembly, the electronic circuitry of the Molded Assembly shall be tested as specified in Appendix A.

3.6 Mine loading. All demonstration mines shall be loaded and packed for final lot acceptance testing specified in the contract.

3.7 Temperature control during molding and cure of molded assemblies. The Molded Assembly shall not have been subjected to temperatures exceeding 165°F at any time when examined as specified in 4.5.8.

3.8 Soldered electrical connections. Soldered electrical connections on the following assemblies shall comply with MIL-S-46844 or MIL-S-45743.

Piston Acutator & Clip Assy	(Dwg. 9366697)
Printed Wiring Board Assembly	(Dwg. 9366692)
Printed Wiring Board Assembly Interconnect	(Dwg. 9298620)
S&A Laminate Assembly	(Dwg. 9366685)
Preform and Sensor Assembly	(Dwg. 9298540)
Preform Assembly	(Dwg. 9366662)
Molded Assembly	(Dwg. 9366661)
Switch Assembly (Terminals)	(Dwg. 9298512)

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3.9 Void criteria for molded assembly. Void with dimension larger than 1/8 inch across by 1/16 inch deep are not allowed except within the large cavity below 1/4 inch from the open end. Within this cavity, voids with dimensions greater than 1/4 inch across by 1/4 inch deep are not allowed. In addition, voids in the S&A and penthouse area of ball cavity are allowed to a maximum dimension of 1/8 inch across by 1/8 inch deep, except the booster pocket must be intact.

Comparison photographs or visual standards may be used to establish visual accept/reject criteria for voids approved by the Contracting Officer.

3.10 Color. The color of the molding compound in the Molded assembly shall be within the range of the colors bounded by colors numbered 34079 through 334102, inclusive of FED-STD-595.

3.11 Mine functions.

3.11.1 Anti-disturbance functioning. The mine shall function in the anti-disturbance mode as the mine is disturbed by two (2) different movements to two (2) different sides.

3.11.2 Breakwire functioning. The mine shall function in the breakwire mode when any deployed tripline is pulled with a minimum force of one hundred and forty grams (140 g) to max force of six hundred grams (600 g).

3.11.3 Self-destruct functioning. The mine shall be capable of self-destructing itself within the self-destruct minus 20% of time limit.

3.11.4 Early functioning. The mine shall not self-destruct itself by any internal mode within 80% of the self-destruct time.

3.11.5 Tripline deployment. The mine after ground impact and with unobstructed field, shall deploy a minimum of three (3) triplines with at least one (1) deployed tripline attaining a minimum deployed length of thread of ten (10) feet and two (2) others providing additional length to total sixteen (16) feet for three (3) triplines.

3.11.6 Premature arming. The mine shall not deploy sensors, fire the piston actuator or gas generator or position the slider in line prior to 25 seconds after depression of the battery ball.

3.12 Duds.

3.12.1 Safe dud. The mine, if not functioned by any external or internal mode, shall not function after the maximum self-destruct time has elapsed.

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3.12.2 Hazardous dud. All safe duds when tested as specified in 4.5, shall not be capable of functioning after the maximum self-destruct time has elapsed.

3.13 Test temperatures and conditioning. The loaded munitions, shall be temperature conditioned for a minimum of four (4) hours prior to testing. Temperature conditioning temperatures are as follows:

a. Minus twenty-five degrees Faharenheit, minus five degrees Fahrenheit (-25°F, -5°F.

b. Seventy degrees Fahrenheit plus or minus ten degrees Fahrenheit (+70°, ± 10°F).

One hundred and forty-five degrees Fahrenheit, plus five degrees Fahrenheit (+145°F, +5°F).

3.14 Munition arming. Arming of demonstration mines will be accomplished by removal of the arming lever which results in battery initiation, shearing of the shorting bar and removal of the S&A safety pin. The Arming Assembly, shall comply with the requirements of Dwg. 12579595 and MIL-A-70780.

3.15 Qualification testing of electrical components. The components of the electronics assembly (P/N 9366692) that require qualification by electrical and mechanical testing are as follows:

9366704	Transistor, Q3
9298496/MIL-R-63993	Resistor Network
9298503	Capacitor, Dual
9298507/MIL-I-48331	Integrated Circuit (CMOS)
9298509	Capacitor C3
9298658	Capacitor C1 and C2
9298555	Transistors, NPN, Q1, Q2
9366694	Capacitor C7 and C8
9366695	Piston acutator
9366691/MIL-STD-55110 and MIL-STD-275	Printed Wiring Board

Components that were qualified in previous contracts for the Molded Assembly do not require requalification except as specified in 3.16.1.

3.16.1 Qualification of components. Qualification of the components listed above shall be accomplished by, recommendation of the contractor for approval by the Government's technical agency.

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3.15.2 Pass/fail criteria. Pass/fail criteria for qualification components will be as follows:

(1) Electrical and mechanical laboratory tests - The components must pass all tests to the specified sampling plan for qualification as defined on the applicable component drawings and specification, otherwise the qualification sample shall be rejected.

3.16 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.4.2.1 through 4.4.2.52, as applicable.

3.17 First article inspection. The specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.18 Workmanship.

3.18.1 Workmanship - demonstration mine. The Demonstration Mine shall be free of dirt, grease, chips, gouges, cracks, and any other defects which would make the item unsuitable for its intended use.

3.18.2 Workmanship - parts and assemblies. Parts and assemblies shall be free from chips, dirt, grease, and other foreign matter that may impair proper functioning. Cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated.

3.18.3 Workmanship - packaging, packing and marking. Packaging and packing materials shall not be water-damaged or contain excessive moisture. They shall not be torn, broken or otherwise damaged, and shall be suitable for the intended use. All marking shall be neat, accurate, and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or qulaity program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

c. Lot Acceptance Testing.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit first article as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below:

a. Certification that each molding compound batch (see 4.5.7.1) used to manufacture the Housing in (c) below complies with the requirements in MIL-M-48245 for compressive modules, flexural strength, flexural modules and area under the flexural strength curve.

b. 18 Molded assemblies (Dwg. 9366661-2) prior to molding over test pads and adding jumper wires.

c. Molded assemblies from (b) above finished as specified on 9366705 (after completion of inspections listed in 4.3.2(c)).

d. One (1) test bar (0.50 x 0.50 x 1.0 inches or 0.50 inch diameter x 1.0 inches long from each molding compound batch (see 4.5.7.1) used to manufacture the Molded assemblies in (b) above. A second test bar shall be made available for the use in the event the first bar fails the test 4.3.2 due to a void. Tolerances on specimen dimensions shall be ± .01 inches

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4.3.2 Inspections to be performed. The First Article parts submitted in accordance with 4.3.2 will be subjected by the Government to any or all the requirements of the applicable drawings and to the examinations, inspections, and tests specified herein.

The test bars for 4.3.1 (2) shall be tested for compressive strength as specified in 4.5.7.2.

b. The 936661-2 Molded Assemblies from 4.3.1 (c) shall be subjected to the following:

1. Upon completion of the tests specified in 4.4.2.27, the jumper wires shall be soldered in and after molding over the test pads the housings shall be subjected to the examinations of 4.4.2.28.

2. The assemblies shall be finished to comply with Dwg. 9366705 for Special Test Mines and shall be subjected to the criteria of 4.4.3.1 through 4.4.3.6.

c. Functioning of mines shall be performed on the 15 mines as per 4.4.3.1. Assembly of Arming Assembly 12579595 is required.

4.3.3 Rejection. The first article quantity shall be rejected if any one of the following occurs:

a. Any S&A Assembly has the transfer lead or lead cup present.

b. The Sensor assemblies subjected to the functioning tests exceed the reject criteria specified in 4.4.2.5.

c. Any anti-disturbance switch fails any of the examinations or tests specified in 4.4.2.7.

d. Any molded assembly fails to comply with any of the electrical tests or examinations specified in 4.4.2.28 and 4.4.2.56.

e. The sample exceeds the allowable defects.

NOTE: The Government reserves the right to terminate its inspecton when any failure of an assembly, component, or test specimen in the sample causes rejection per the accept/reject criteria for that examination or test.

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Table I. First article testing

	Cold (-25°F + 5°F)	Ambient (70° ± 5°F (145 ± 5°)	Hot
Breakwire	2	2	2
Anti-disturbance	2	2	2
Self destruct	2/6	2/6	2/6

Total 18

First article and production - (sample size of 18 test mines. The lot shall be rejected if three or more test mines fail to comply with any of the requirements specified in 3.4, and 3.11. 18-2-3).

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4.4 Quality conformance inspection.

4. 4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Inspection lots of Housing, Timing and Fuzing shall contain.

a. Electrical detonators (Dwg. 9245691) of one lot interfix number from one manufacturer (see 4.5.3).

b. Battery assembly (MIL-B-70731) from one manufacturer.

c. Gas generator (Dwg. 9298530) of one lot interfix number from one manufacturer.

d. Integrated circuits (Dwg. 9298507) from one manufacturer.

e. Thick film resistor network (Dwg. 9298496) from one manufacturer.

f. Darlington transistor (metal can) shall be from one manufacturer and of one lot interfix number.

4.4.2 Lot acceptance testing. Shall be performed as per 4.4.3.1.

4.4.3 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Testing, sampling plans for major and minor defects shall be in accordance with MIL-STD-105. Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Printed Wiring Board, Electronics	1 OF 1	9366692
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Pink Board		100%	3.2	Visual
102	Printed Wiring Board Material			3.1	Certification
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:
1. Sample in accordance with MIL-STD-105D, Inspection Level II. Acceptance of the lot is contingent upon finding zero (0) defects in the sample.

AVSCOM Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR).

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298547
4.4.2.2	Manifold Assembly, Center				NEXT HIGHER ASSEMBLY 9298540
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any passage blocked		0.40%	3.2	Test
102	Tubing punctured, cut, loose or improperly assembled		0.40%	3.2	Visual
103	Manifold tubing over ignitor (gas generator) not heat shrunk		0.40%	3.2	Visual
104	Gas generator assembly missing		0.40%	3.2	Visual
105	Ball missing or loose		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Manifold Assembly, Side				9298543
					NEXT HIGHER ASSEMBLY 9298540
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any passage blocked		0.40%	3.2	Test
102	Ball missing or loose		0.40%	3.2	Visual
103	Tubing punctured, cut, loose or improperly assembled		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Diaphragm	1	1	9298598
				NEXT HIGHER ASSEMBLY 9298598

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of tears or cracks after mandrel test		0.40%	3.2	Visual
102	Material	NOTE 1			Certification
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. Certification of compliance for material shall include the 100% virgin material requirement.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Sensor Assembly				9298569
					NEXT HIGHER ASSEMBLY 9298540
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Functioning Tests (NOTE 1)				
101	Non-release (NOTE 2)		0.40%		
102	Release and thread security		80-8-9	3.4.1	4.5.4.1
103	Tripline deployment		80-5-6	3.4.2	4.5.4.2
104	Breakwire functioning		80-8-9	3.4.3	4.5.4.3
105	Post retention and thread breaking strength		80-7-8	3.4.4	4.5.4.4
106	Solder connections improper (NOTE 3)			3.8	3.8
107	Post locking ball missing (NOTE 2)		4.0%	3.2	X-ray
108	Sensor intrusion test		80-8-9	3.4.5	4.5.4.6
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:
1. The tests shall be performed in the order listed. The sample units for each test shall have been tested on all preceding tests in this sequence. However, acceptance or rejection on one test shall not constitute acceptance or rejection on the other tests. The sample sizes specified are applicable to each test and shall not be amended by fallout from previous tests.
2. This test to be performed prior to assembly of the breakwire, breakwire retaining ring, switch post cover and epoxy. The testing may be conducted as a normal production operation with an inspector verifying the operation.
3. Sample size shall be in accordance with appendix B.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298515
4.4.2.6	Can, Switch				NEXT HIGHER ASSEMBLY 9298514
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Plating	5 per batch		3.2	4.5.5.3
Minor 201	Poor workmanship		1.0	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-OA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Switch Assembly	1 OF 1	9298512
			NEXT HIGHER ASSEMBLY 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch function seven (7) contact closures minimum each direction		100%	3.2	4.5.5.1
102	Contact closure duration 0.60 seconds maximum		100%	3.2	4.5.5.1
103	Insulation resistance		100%	3.2	4.5.5.2
104	Assembly damaged, distorted or incomplete		0.40%	3.2	Visual
105	Ribbon improperly welded to can		0.40%	3.2	Visual
106	Epoxy seal missing or incomplete		0.40%	3.2	Visual
107	Solderability, 2 leads (NOTE 1)			3.8	Visual (NOTE 1)
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: Verify double dip operation each shift.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Printed Wire Board Assembly Interconnect	1 OF 1	9298620
			NEXT HIGHER ASSEMBLY 9298451

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Natural color board without hole in shorting bar		100%	3.2	Visual
102	Ribbons not welded to battery		100%	3.2	Visual
103	Shorting bar missing		100%	3.2	Visual
104	Ribbons not soldered to printed wiring board		100%	3.2	Visual
105	Battery, Battery flange insulator on black plug missing		0.40%	3.2	Visual
106	Solder workmanship (NOTES 1 and 2)		0.40%	3.8	Visual
107	Weld pull test				4.5.6
108	Printed wiring board material			3.1	Certification
Minor				3.8	Visual
201	Poor workmanship		1.0%	3.18	Visual

NOTES:
1. The AQL is applied to the number of boards in the lot.
2. Accept/reject criteria in accordance with Appendix B, para. 50.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Dual Transistor, NPN Darlington	1 OF 1	9298555
			NEXT HIGHER ASSEMBLY: 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Temperature Cycling (NOTE 1)				
101	Forward-current transfer ratio, HFE @ 85°C		100%	3.2	MIL-STD-883/Method 1010/Condition C
102	Collector-emitter cut off current, ices @ 85°C		100%	3.2	MIL-STD-750/Method 3076
103	Base-emitter voltage, VBEF @ 85°C		100%	3.2	MIL-STD-750/Method 3041
104	Collector-emitter saturation voltage, VCES @ 85°C				MIL-STD-750/Method 3066
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. Test to be performed at 85°C after subjecting the components to twenty-five (25) cycles of temperature shock. 2. Only those transistors successfully completing the above tests shall be used as Q1 (Detonator circuit) in 9298495 electronics assembly.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Capacitor, Tantalum, Dual, Dip Case	1 OF 1	9298503
			NEXT HIGHER ASSEMBLY 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Thermal Shock (NOTE 1)				
101	Capacitance		0.40%	3.2	NOTE 4b/Dwg. 9298503
102	Dissipation Factor		0.40%	3.2	NOTE 2a/Dwg. 9298503
103	DC Leakage (Ambient Temperature)		0.40%	3.2	NOTE 2c/Dwg. 9298503
104	DC Leakage (Plus 85°C		0.40%	3.2	NOTE 2d/Dwg. 9298503
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with NOTE 4b of Dwg. 9298503.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Capacitor, Fixed, Tantalum, Solid, Polarized	1 OF 1	9298509
			NEXT HIGHER ASSEMBLY: 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Thermal Shock (NOTE 1)				Method 107, Cond. A, MIL-STD-202
101	Capacitance		0.40%	3.2	NOTE 3b1/Dwg. 9298509
102	Dissipation Factor		0.40%	3.2	Method 305/MIL-STD-202 NOTE 3b2/Dwg. 9298509
103	DC Leakage at 6VDCm (4 US maximum)		0.40%	3.2	Method 306/MIL-STD-202 NOTE 3b3/Swg.9298509
Minor 201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with Method 107, Test condition A, MIL-STD-202.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.12	Capacitor, Fixed Plastic Dielectric	1	1	9298510
				NEXT HIGHER ASSEMBLY 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	Thermal Shock (NOTE 1)				NOTE 4b/Swg. 9298510
101	Capacitance		0.40%	3.2	NOTE 2a/Dwg. 9298510
102	Dissipation Factor		0.40%	3.2	NOTE 2c/Dwg. 9298510
103	Insulation resistance (Ambient Temp.)		0.40%	3.2	NOTE 2d/Dwg. 9298510
104	Insulation resistance (Plus 85°C)		0.40%	3.2	NOTE 2d/Dwg. 9298510
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. Tests to be performed after subjecting components to five (5) cycles of thermal shock in accordance with NOTE 4b of Dwg. 9298510.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298503
4.4.2.13	Resistor, Network, Thickfilm				NEXT HIGHER ASSEMBLY 9366692
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of Dwg. 28106068			3.2	Certification
Minor 201	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.14	Integrated Circuit, Control and Timing Base CMOS	1	1	MIL-I-48331/9298507
				NEXT HIGHER ASSEMBLY: 9366692

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnish evidence of having satisfactorily met inspection requirements of MIL-I-48331			3.2	Certification
Minor 201	None defined				

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH 4.4.2.15	TITLE Printed Wiring Board Assembly		SHEET 1 OF 1		DRAWING NUMBER 9366692
					NEXT HIGHER ASSEMBLY 9298490
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire connections improper		0.40%	3.2	Visual
102	Solder workmanship (NOTES 1 and 2)		0.40%	3.2	Visual
103	Land paths broken, raised or bridged with solder		0.40%	3.2	Visual
104	Printed wiring board cracked or broken		0.40%	3.2	Visual
105	Components damaged or missing		0.40%	3.2	Visual
106	Components mounted with wrong polarity		0.40%	3.2	Visual
Minor					
201	Maximum height		0.65%	3.2	Gage
202	Poor workmanship		1.0%	3.18	Visual

NOTES: The AQL is applied to number of the boards for each lot submitted. 2. Accept/reject criteria in accordance with Appendix B, para 10.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9366685-2
4.4.2.16	S&A and Laminate Assembly				NEXT HIGHER ASSEMBLY 9366662-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Sealant or foreign material on exposed surface of lead cup assembly		0.40%	3.2	Visual
102	Soldered connections improper		0.40%	3.2	Visual
103	Shorting washer not removed		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	S&A Assembly, PDM		9366668-2
			NEXT HIGHER ASSEMBLY 9366685-2

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Presence of transfer lead		100%	3.2	Visual
2	Presence of acceptor lead		100%	3.2	Visual
Special					
1	NOTE 3 Presence of shorting device		100%		Visual
Major					
101	Presence of parts				
	a. Sleeve, S&A 9366683 per NOTE 6		100%		Visual
	b. Seal, washer 9298459-1 per NOTE 2		0.40%		Visual
	c. Gasket - 9366667		0.40%		Visual
	d. Seal, washer 9298459-2		0.40%		Visual
	e. Clip, terminal		0.40%		Visual
	f. Det., electric, M100 9245691		0.40%		Visual
	g. NOTE 5 adhesive/sealant		0.40%		Visual
Minor					
201	Poor workmanship		0.65%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Sleeve, S&A	1 OF 1	9366683
			NEXT HIGHER ASSEMBLY: 9366668

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material verfification per NOTE 2		1-0-1		Lab testing
102	Thickness		100%		Gage
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Clip, Terminal				9298460-2
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of material		0.40%	3.3	Gage
102	ID of 2 clip		0.40%	3.3	Gage
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTE

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Wire, Housing				9366686
					NEXT HIGHER ASSEMBLY 9366674
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Material verification per NOTE 2		1-0-1		Lab testing
Minor 201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	Preform Assembly				9366662-2
					NEXT HIGHER ASSEMBLY 9366661-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldered connections improper or unsoldered		0.40%	3.2	Visual
102	Retainer and ball assembly not seated on battery		0.40%	3.2	Visual
103	Any component or assembly not secured to preform		0.40%	3.2	Visual
104	Kinked, broken, loose or flattened sensor tubing		0.40%	3.2	Visual
105	Any unconnected sensor tubing		0.40%	3.2	Visual
106	Any wire trimmings present		0.40%	3.2	Visual
107	Presence of solder bridge		0.40%	3.2	Visual (NOTE 1)
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES: 1. For safety purposes, soldering bridge must be present on part from time detonator leads are connected until just prior to conduct of detonator resistance test (see 4.5.12).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9366661-2
4.4.2.22	Molded Assembly				NEXT HIGHER ASSEMBLY 9366700 or 9366705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety delay (less than 25 seconds)		100	3.2	Gage
Major					
101	Jumper wires not soldered to pads		0.40%	3.2	Visual
102	Solder connections improper		0.40%	3.8	Visual
103	Temperature control during cure and cure time (NOTE 1)			3.7	4.5.8
104	Resin/Hardener weight ratio of molding compound		0.40%	3.2	4.5.7.2
105	Shelf life of molding compound expired		0.40%	3.2	Visual
106	Functional testing of electronic circuitry		0.40%	3.5	4.5.9
107	Any unconnected sensor wires (NOTE 2)		0.40%	3.2	Visual
108	Removal of solder bridge		1.0%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:
1. Cure temperature and cure time shall be monitored for each molding compound batch of housings.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.23	Molded Assembly				9366661-2
					NEXT HIGHER ASSEMBLY 9366700 or 9366705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special 1	Shorting bar broken or damaged		100%	3.2	Visual or approved gages
Major 101	Presence of bare (uninsulated) or broken wire in cavity		0.40%	3.2	Visual
102	Voids in excess of specified		0.40%	3.9	Visual
103	Detonator sleeve missing or loose		0.40%	3.2	Visual
104	Any sensor locating plug missing (7 places)		0.40%	3.2	Visual
105	Molding material or other foreign material on exposed surface of lead cup assembly over 0.005		0.40%	3.19	Visual
106	Incomplete fill in test pad area		0.40%	3.2	Visual
107	Temperature control during cure minimum cure time		NOTE 1	3.7	NOTE 1
108	Weight ratio of molding compound		NOTE 1	3.7	NOTE 1

NOTE 1: Cure temperature, cure time and weight ratio shall be monitored for each batch housings.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.28	Molded Assembly				9366661-1
					NEXT HIGHER ASSEMBLY 9366700 or 9366705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major	(Continued)				
109	Distance from base of largest cavity diameter to apex gaging surfae maximum		0.40%	3.2	Gage
110	Color incorrect		0.40%	3.10	Visual
111	Distance from end face to forward edge of shorting bar		0.40%	3.2	Gage
112	Height or bare battery ball above apex radius		0.40%	3.2	Gage
113	Height of shorting bar above apex radius		0.40%	3.2	Gage
114	Max distance from arc face (center area) to apex gaging radius		0.40%	3.2	Gage
115	Battery ball free of potting material over designated area		0.40%	3.2	Gage
116	Functional test of electronics		100%	3.5	4.5.9
117	Temporary safety pin assembly missing or defective		0.40%	3.2	Visual
Minor					
201	Plug under shorting bar removed		0.65%	3.2	Gage
202	Molding material in dovetail of S&A or booster slot		0.65%	3.2	Gage
203	Loose end face laminates		0.65%	3.2	Visual/Manual
204	Any exposed tubing which is twisted, kinked, punctured or broken		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9366661-1
4.4.2.25	Molded Assembly				NEXT HIGHER ASSEMBLY 9366700 or 9366705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor	(Continued)				
205	Surface preparation of largest diameter of cavity missing or incomplete		0.65%	3.2	Visual
206	Any markings missing, incorrect or incomplete		0.65%	3.2	Visual
207	Poor workmanship		1.0%	3.18	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Mine Assembly Demonstrator, PDM				9366698
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Presence of kill mechanism		100%	3.2	
2	KM cavity molded to top		100%	3.2	
Major					
101	Presence of parts				
	a. Arming assembly 12569595		0.40%		Visual
	b. 2-screw, panhead - MS51957-14		0.40%		Visual
	c. Screw, tapping, panhead - MS24630-12		0.40%		Visual
	d. Hook, shorting bar 12579585		0.40%		Visual
	e. Clip, shorting bar 12579586		0.40%		Visual
102	Adhesive per NOTE 5		0.40%		Visual
103	Screw torque per NOTE 3		0.40%		Torque wrench
104	Insulator placement per NOTE 10		0.40%		Visual
105	Warning notice improper or missing on arming straps		0.40%		Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Bag Assembly Demonstration Mine for M86	1 OF 1	12910119
			NEXT HIGHER ASSEMBLY: 12910123

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Wrong model mine inserted		100%	3.2	Visual
Major					
101	One or more tear tabs missing or incorrectly formed		0.40%	3.2	Visual
102	Heat seal incorrectly made		0.40%	3.2	Visual
103	Pull tab positioned incorrectly		0.40%	3.2	Visual
Minor					
201	Heat seal positioned incorrectly		1.0%	3.2	Gage
202	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Demonstration Mine with Bandoleer for M86 APERS Mine	1 OF 1	12910123
			NEXT HIGHER ASSEMBLY: Final

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Wrong model mine inserted		100%	3.2	Visual
Major					
101	Mine not loaded correctly		0.40%	3.2	Visual
102	Marking for bandoleer defective or positioned incorrectly		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.29	Packing and Marking for Shipping & storage container, PA 19 for Mine, APERS M86, Classroom model, Special test Mine, Demonstration Mine	1	1	9366709
				NEXT HIGHER ASSEMBLY: 9366710

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
1	Marking on PA19 container incorrect		100%	3.2	Visual
Major					
101	Packing material(s) incorrect sizes		0.40%	3.2	Visual
102	Moisture in packing material above max.		0.40%	3.2	Gage
103	Outer barrier bag marked incorrectly		0.40%	3.2	Visual
104	Mine inserted incorrectly into inner bag and bandoleer		0.40%	3.2	Visual
105	Outer bag loaded incorrectly		0.40%	3.2	Visual
106	Outer bag sealed incorrectly		0.40%	3.2	Visual
107	Pull tabs missing from outer bag		0.40%	3.2	Visual
108	PA19 container not loaded correctly		0.40%	3.2	Visual
109	Marking for outer bag incorrect		0.40%	3.2	Visual
110	Marking for PA19 container incorrect		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-70779 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Packing and Marking for Box Wirebound for Mine, APERS M86, Classroom model, Special test Mine, Demonstration Mine	1 OF 1	9366710
			NEXT HIGHER ASSEMBLY: 9366711

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packing material missing		0.40%	3.2	Visual
102	Incorrect packing material used		0.40%	3.2	Visual
103	Moisture content of packing materials over max.		0.40%	3.2	Gage
104	Packing material installed incorrectly		0.40%	3.2	Visual
105	Marking incorrect or missing		0.40%	3.2	Visual
106	Wirebound box closed incorrectly		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.18	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.3.1 Lot acceptance testing. Eighteen (18) - QTY Demo per lot shall be used for lot acceptance testing. The rounds shall be conditioned and tested as follows:

Temperature	Quantity
-25°F - 5°F	6
+70°F ± 10°F	6
+145°F + 5°F	6

4.4.3.2 Mine test distribution. Each test lot of munitions shall be divided so that approximately one-third (1/3) of the mines are tested for anti-disturbance, one-third (1/3) of the mines are tested for breakwire and one-third (1/3) of the mines are tested for self-destruct.

4.4.3.3 Self-destruct functioning. The lot shall be rejected if any of the self-destruct functions exceed any of the defect quantities as follows:

Safe duds: (3%)
Early outs: (3%)

4.4.3.4 Anti-disturbance functioning. Mines designated for anti-disturbance testing shall be tested for the specified criteria in accordance with 4.5.12.1. Also, mines-failing to function during the first breakwire function attempt in accordance with 4.5.12.2 and subsequently tested for anti-disturbance functioning test observed providing the sample is not classed as a safe dud. The results of mines undergoing a second breakwire function attempt shall be for information only and not included in lot acceptance.

The lot shall be rejected if more than 6% of the mines tested for anti-disturbance functioning are classed defective.

4.4.3.5 Breakwire functioing. Mines designated for breakwire testing shall be tested for the specified breakwire criteria in accordance with 4.5.12.2. Also, mines failing to function in the anti-disturbance mode in accordance with 4.5.11.1 shall be tested for breakwire functioning in accordance with 4.5.11.2. Mines which fail to deploy any triplines are to be scored as a no test for breakwire mode and deleted from the sample.

Mines either functioning or failing to function on the first breakwire attempt shall be included in the number of breakwire functioning test observed providing the sample is not classed as a safe dud.

The results of mines underdoing a second breakwire attempt shall be recorded for informational purposes only.

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The lot shall be rejected if more than six percent (6) of the mines tested for breakwire functioning is classed defective.

Mines failing to function on the first breakwire attempt shall be tested for anti-disturbance functioning in accordance with 4.5.12.1 as noted in 4.4.3.4.

4.4.3.6 Tripline deployment. All mines shall be visually examined for triplines deployed and, where required, for length of lead deployed. The results of the above shall be recorded. The lot shall be rejected if more than six percent (6%) of all mines conditioned to +70°F are classed defective. Tripline deployment for mines conditioned at the temperature extremes shall be recorded.

4.4.3.7 Premature functioning. All functioning mines shall be tested for premature functioning. Depressing the battery ball with the CAM determines the initiation of the arming cycle. The requirement is that no functioning (i.e. gas generator or detonator) for a minimum of 25 seconds after initiation of the arming cycle.

4.4.3.8 Data acquisition. All data pertaining to mine testing shall be recorded.

Included with lot acceptance criteria, records shall be maintained of the number of non-hazardous mines which function early or are complete duds as well as approximate time of early functioning.

4.4.3.9 Sensor intrusion test. Each lot of sensors shall be tested for the epoxy intrusion requirement (defined by 3.4.5 and 4.5.4.6) to determine if there is any epoxy intrusion. Testing for this requirement shall be completed prior to lot acceptance testing for sensor functional requirements (as defined in 3.4.1 through 3.4.4) . The sampling plan used for the intrusion test defined in 3.4.5 shall be 80-8-9. Any lot failing to comply with the sampling shall be rejected.

4.4.3.10 Sensor wire continuity. As an alternate to inspecting a sample of the parts, the contractor may perform an inspection to verify the effectiveness of a 100% production screen for continuity of the sensor circuits. The inspection must verify that the test equipment and production operator are making the correct decision during the 100% screening operation. The plan for this verification must be documented and forwarded to the Technical Agency for approval (see 6.4).

4.4.4 Inspection equipment. The inspection equipment required to perform the examinatios and tests prescribed herein is described in the table beginning with 4.4.2.1. The contractor shall submit, for approval, inspection equipment designs in accordance with the terms of the contract.

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4.5 Methods of inspection.

4.5.1 Arming of S&A assembly. After removing the safety pin from thirty (30) of the S&A assemblies, the piston actuator shall be initiated, Standard safety practices shall be employed. The slider should move to the full-armed position and the M100 fired. A success will constitute a propagation from the M100 thru the acceptor cup, transfer lead up to and including the lead cup.

4.5.2. Non-arming of S&A assembly. Thirty (30) S&A assemblies shall be tested for non-arming by initiating the piston actuator. The safety pin must remain intact, blocking movement of the slider. Standard safety practices shall be employed. The S&A assemblies in 4.5.1 and 4.5.2 shall be molded in test blocks. The design of the test blocks shall be approved by the Government technical agency prior to testing.

4.5.3 Qualification test of electric detonator, transfer lead assembly and lead cup assembly lots. This qualification test shall be performed for the combination of detonator serial lot, transfer lead assembly and lead cup assembly serial lot. The components to be used for this test shall be randomly selected from the respective lots. All detonators selected for this test shall remain shorted, (leads twisted) until immediately prior to firing. The components shall be assembled into S&A's comprised of housing, slider assembly, gasket and detonator side plate. (Piston actuators are not required.) The S&A's shall meet applicable drawing requirements, including those affecting the gaps between the detonator, transfer lead assembly and lead cup assembly. Detonating cord assemblies form an accepted lot shall be suitably attached next to the lead cup assembly with a .020 inch minimum spacing between the two assemblies. The short shall then be removed and the detonators shall be initiated from behind an approved barricade using a suitable power source (a 100 microforad capacitor charged to 1.6 volts).

Observation shall be made for failure to initiate the detonating cord assembly. Approved barricades shall be used for assembling and testing these components. If the number of assemblies failing to initiate the detonating cord exceeds the specified acceptance criteria, that lot combination of detonators, transfer lead assemblies and lead cup assemblies shall be rejected for use in the housing, timing and fuzing.

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4.5.4 Sensor assembly.

4.5.4.1 Release and thread security of sensor assembly. The breakwire resistance shall be checked for continuity. Units which have resistance greater than 1,000 ohms shall not be subjected to this test. The units to be tested shall be placed in a suitable fixture and the pressure specified in 3.4.1 shall be applied. A unit shall be classified defective if any of the following occur:

a. Fails to release when initially pressurized as evidenced by the bobbin core being retained within the sensor case.

b. The breakwire continuity is broken as evidenced by a resistance greater than 1,000 ohms across the exposed ends for the breakwire.

c. The thread does not remain attached to the bobbin and post.

d. The thread breaks during release.

e. Any unit that does not release when initially pressurized, but released when the pressure is reapplied, shall be classed defective. This test is destructive. Those units capable of meeting the requirements of the test specified in 4.5.4.2 shall be used for that test even if they failed to meet the requirements for this test.

4.5.4.2 Tripline deployment of sensor assembly. After the sensor assembly has been tested in accordance with 4.5.4.1, and with the sensor remaining in the test fixture, observation shall be made to assure that the bobbin core, with tripline attached, is deployed over a vertical barrier as follows:

a. The barrier location shall be 16 feet, minimum, measured horizontally from the sensor.

b. Barrier height shall be 7 inches, minimum, above the top of the sensor. A bobbin core that impacts ground level before the barrier or impacts the barrier shall be considered defective. This test is destructive.

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4.5.4.3 Breakwire function of sensor assembly. After the sensor assembly is tested in accordance with 4.5.4.1, 4.5.4.2 and with the unit remaining in the test fixture, a suitable force gage shall be attached to the tripline at a point just above the barrier. A pull force shall be applied along the axis of the bobbin post and the breaking force shall be noted. The breaking force is that force at which the resistance across the exposed ends meets the requirements specified in 3.4.3. Any unit that fails the pull force test, as specified in this paragraph, shall be classified defective. This test is destructive. Those units that are capable of meeting the requirements of the test specified in 4.5.4.4 shall be used for that test even if they failed to meet the requirements of this test. The rate of pull shall be a person walking at a normal rate.

4.5.4.4 Post retention and thread breaking strength. After the sensor assembly is tested in accordance with 4.5.4.1, 4.5.4.2, 4.5.4.3 and with the unit remaining in the test fixture, the specified force shall be applied at rate of 4 inches/second minimum to the tripline along the axis of the bobbin post. Any unit that releases the bobbin post or on which the tripline breaks shall be classed defective. This test is destructive. Units so tssted shall not be returned to the lot. The rate of pull shall be a person walking at a normal rate.

4.5.4.5 Soldered electrical connections of sensor assembly. The electrical connections of the breakwire to the breakwire terminals shall be tested for electrical continuity using a suitable ohmmeter, and shall be tested for mechanical bond strength using a suitable force gage. The force shall be applied at the center of the breakwire through a hook or similar fixture that does not damage the wire at the point of application. The force shall be applied perpendicular to the breakwire terminal and in the direction that will lift the breakwire from the terminal.

Any assembly which fails the electrical continuity test or the mechanical bond strength test shall be classed defective.

The mechanical bond strength test is destructive. Assemblies so tested shall not be returned to the lot.

4.5.4.6 Molding material intrusion test. The sensor assembly will be placed into a test brick mold with twelve (12) sensors per mold. Shrink tubing will be assembled to the sensor nozzles and exit the mold. Sensor plugs will be assembled to the cap end. The mold will be closed and subjected to the normal wedge molding process. The test brick cure will be for a minimum of 16 hours at 155° + 10°F cure temperature. The sensor caps shall be removed and the test brick assembled to an approved test fixture. The pressure specified in 3.4.5 shall be applied to the protruding end of the shrink tubing. This is a destructive test. Units may not be returned to the lot.

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4.5.5 Anti-disturbance switch.

4.5.5.1 Anti-disturbance switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 3600 on its cylindrical axis, while this axis is held at 15° ± 2° referenced to the horizontal, for both clockwise and counter-clockwise directions of rotation. The functional requirements of drawing 9298512 shall be met.

This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.5.2 Insulation resistance of anti-disturbance switch. The switch assembly, when mounted in a suitable fixture, shall be measured for insulation resistance in the rest position using an approved megohmmeter. The megohmmeter reading shall meet the requirements of drawing 9298512. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.5.3 Plating of anti-disturbance switch components. Five (5) components shall be randomly selected form each plating batch. The specimens shall be checked in accordance with MIL-G-45204. If any sample fails to comply with the plating requirements, the sample shall be classed defective (Destructive test).

4.5.6 Weld test of printed wiring board assembly, interconnect. A sample of one (1) part from the first ten (10) welded shall be tested to qualify the welding equipment for use in production. in addition, one (1) sample shall be tested after production of one each one-hundred (100) welded parts as a control of welding equipment during production. The units shall be placed in a fixture capable of applying the force specified on the applicable drawing. Any assembly in which the weld cracks or breaks when the specified force is applied shall be classed defective.

If any defective parts are found during qualification of welding equipment the equipment producing the defective assembly shall not-be used in production.

If any defective assembly is found when inspection is performed for control of the welding equipment, the equipment producing the defective weld shall be removed from production. Further, that portion of production since the last weld control check shall be returned to the contractor for corrective action.

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If welding is stopped for one (1) hour or longer, the equipment shall be subjected to the qualification test prior to use in production.

All welding equipment removed from production because of some fault may, after reworking, be returned to production, providing it passes the qualification test above.

The contractor may use coupons of the same material and thickness as the parts being welded subject to approval of the Contracting Officer.

4.5.7 Qualtiy controls for each molding compound batch for housing or plug.

4.5.7.1 Resin/hardener weight ratio control. At the beginning of production and at least once each four hours of production, or portion thereof, the inspector shall draw two (2) test shots with the mixer set to expel the hardener and resin separately. The hardener and resin of each shot shall be weighed individually to verify that the ratio is between 1.8/1 and 2.2/1. If either shot fails to meet the requirement, all items filled since the last acceptable check shall be rejected. Further, the mixer shall be corrected and shall pass the above test before being returned to use.

4.5.7.2 Mixing and cure control. One (1) test bar from each batch of molding compound (see 4.5.7.1) shall be molded at the same time the batch is used to mold housings or plugs, or bond plugs to housings for lot acceptance testing. Vibration and evacuation as necessary may be used to mold the bar. The mix ratio limits shall be 2.0/1.0 + 10% (1.8/1 to 2.2/1) . The test bar shall be processed with the items molded from the batch of molding compound which the bar represents; i.e., the bar shall be cured in the same ovens and subjected to the same temperatures for the same periods as the time it represents. After completion of the cure,- the bar shall be tested for ultimate compressive strength, as specified in ASTM-D-695(85) except that the specimen dimensions shall be 0.5 inch by 0.5 inch by 1.0 inch or 0.5 inch diameter by 1.0 inch long. The ends shall be flat and parallel to within .003 inch, and shall be perpendicular to the long axis of the specimen. Speed of testing shall be as specified in the ASTM.

4.5.8 Temperature control during molding/cure operations. Whenever the preform assembly, the housing, or the munition is subjected to heating in ovens, temperature records shall be maintained and reviewed to assure that the minimum temperature has been reached and that the maximum temperature has not been exceeded. If any units have been subjected to temperatures beyond the maximum limit at anytime, regardless of duration, they shall be rejected. Temperature measuring instruments shall be approved and calibrated in-accordance with 4.4.4.

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4.5.9 Functioning of the electronic circuity of the molded assembly. The electronic circuitry of each housing in the lot shall be tested after molding and prior to filling the test conductor areas. Each housing shall be subjected to the tests listed below in accordance with Appendix A. All housings containing any defect shall be removed from the lot.

Test	Paragraph	Limits
Detonator and Gas Generator Logic	40.2.4	Improper
Preliminary Oscillator Frequency	40.2.1	a) Less than 97% of nominal b) More than 104% of nominal c) Equal to or greater than 97% and less than 99% of nominal or greater than 102% but less than or equal to 104% of nominal.
Barrier Oscillator Frequency	40.2.1	a) Less than 97% of nominal . b) More than 104% of nominal. c) Equal to or greater than 97% and less than 99% of nominal or greater than 102% but less than or equal to 104% of nominal.
System Operating Current	40.2.2	Greater than 1.8 milliamps but less than 2.40 milliamps.
Self-Destruct Time Out	40.2.3	a) Greater than 100.1% of nominal

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Low Voltage	40.2.5 At TA=+25°C +5°C (+77°± ±10°)	a) Type I: Non-functioning, or functioning less than. b) Type I: Equal to or greater than 3.2 volts but less than 3.4 volts.
Detonator Resistance	40.2.9 (see note)	Less than 2.9 ohms greater than 9.5 ohms.

NOTE: This test shall be 100% performed on Part #9298451 prior to molding, and precaution shaold be taken to assure that the solde bridge is present to prevent inadvertent functioning of the detonator during handling prior to test. Assurance shall be provided that the solder bridge is removed immediately prior to the resistance test so as not to invalidate test results.

Test	Paragraph	Limits
Output Capacitor Voltage	40.2.6	Improper
Continuity	40.2.12	a) Both circuits open. b) One circuit open.
Anti-disturbance Switch Operation	40.2.14	

The following tests shall be performed after adding the permanent detonator short.

Detonator Short	40.2.11	Absence of Short
Gas Generator II Piston Actuator Resistance	40.2.10	a) Equal to or less than 1.1 ohms or equal to or greater than 2.9 ohms. b) Greater than 4.0 ohms but less than 8.0 ohms.

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4.5.10 Pull test. The sample shall be fixtured and the ribbon lead shall be pulled in the specified direction to meet the specified minimum force. The sample shall not indicate failure of the weld during the test. Any sample failing to comply with all of the specified requirements shall be classed defective (Destructive test).

4.5.11 Mine functioning.

4. 5.11.1 Anti-disturbance functioning. The sample selected for anti-disturbance functioning shall be tested in the following manner:

a. The sample to be tested shall be moved in any of the specified directions, depending on ground orientation.

b. All data shall be recorded.

c. If the mine fails to function it shall be tested for breakwire activation in accordance with 4.5.12.1(a) through 4.5.12.2(e).

d. If the mine fails to function self-destruct it shall be tested in accordance with 4.5.14 for the hazardous dud requirements of 3.13.2 (Destructive test).

4.5.11.2 Breakwire functioning. The sample selected for breakwire functioning shall be tested in the following manner:

a. The sample to be tested shall be restricted from any movement.

b. Any tipline from either breakwire circuit shall be selected for an activation attempt.

c. A digital readout force gage shall be attached to the lead and an axial force shall then be applied to the device under test.

d. The tripline, with gage attached, shall be pulled until either the mine functions or the triplines fails to function within the specified limits.

e. All data shall be recorded. The mine shall be classed defective if it fails to function with the specified force criteria for breakwire functioning.

In the event the mine fails to fire on the first breakwire attempt, the following procedure shall be followed:

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f. The mine shall be tested for anti-disturbance functioning in accordance with the procedures of 4.5.12.1 and the requirements of 3.11.1 and shall be scored.

g. If the mine fails to function in the anti-disturbance mode, another tripline form the adjacent breakwire circuit shall be tested as noted above for breakwire functioning and shall be scored.

h. If the mine fails to function in the second breakwire attempt, it shall be left to time out for self-destruct functioning.

i. If the mine fails to function self-destruct, it shall be tested in accordance with 4.5.14 for the hazardous dud requirements of 3.13.2 (Destructive test).

4.5.11.3 Self-destruct function. The mines selected for self-destruct functioning shall be flagged and monitored for detonation times. All times of detonation, early or normal, shall be recorded (Destructive test).

4.5.12 Tripline deployment. All samples shall be visually examined and measured when required for the tripline deployment criteria of 3.12. Any mine failing to meet the specified criteria shall be classed defective.

4.5.13 Hazardous dud evaluation procedure. (Non-hazardous mine) Any mine which has failed to function in the breakwire mode, anti-disturbance mode or failed to function self-destruct after the maximum time out has elapsed shall be subjected to the following inputs to determine if the detonator can be fired:

a. One additional tripline, on the item, shall be pulled. All other triplines shall be cut so as not to activate the remaining breakwires.

b. The item shall be tipped about the arc face, to the slant face, to the apex and to the opposite slant face.

c. Repeat b.

The inputs shall be applied in the order listed above. If the detonator functions when any of these inputs is applied, the unit shall be classed as a hazardous dud. If the detonator does not fire all these inputs are applied, it shall be classed as a non-hazardous dud.

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4.5.14 Slider integrity. The contractor shall submit to the government a test plan for testing the integrity of each sintered slider lot. As a minimum three temperatures must be used for testing (-45°±5°F, 70±5°F, 145±5°F). A portion of the test plan must include testing–the loaded slider P/N 9366684 at all three temperatures with an M100 firing directly into the wall of the loaded slider. The test quantity shall be no less than 10 pieces at each temperature (30 at cold temp) for a total of 50 loaded sliders being fired on. A failure (critical) will be assumed if evidence of any cracking or spalling of the slider takes place or an initiation of the loaded slider occurs. Submission of the test procedure shall be submitted to AMSMC-QAT-M for government approval prior to testing.

5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Level A unit packaging shall be in accordance with Dwg. 9366709.

5.1.2 Level B. Level B unit packaging shall have the same requirements as for Level A.

5.1.3 Level C. Level C unit packaging shall have the same requirements as for Level A.

5.2 Packing.

5.2.1 Level A. Level A packing shall be in accordance with Dwg. 9366710.

5.2.2 Level B. Level B packing shall have the same requirements as for Level A.

5.2.3 Level C. Level C packing shall have the same requirements as for Level A.

5.3 Marking.

5.3.1 Level A. Level A marking shall be in accordance with Dwg. 9366710 and Dwg. 9355710.

5.3.2 Level B. Level B marking shall have the same requirements as for Level A.

5.3.3 Level C. Level C marking shall have the same requirements as for Level A.

5.4 Shipping. When shipments of units from more than one lot are made at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Intended use. the assemblies covered by this specification are intended for use in a Mine, Anti-Personnel, M86.

6.2 Ordering data. See MIL-A-48078 (PA).

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, USARDEC, ATTN: AMSMC-QAT-M (D) , Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

Request that letter of submittal state the contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Molding compound batch definition. A batch of molding compound is herein defined as that quantity of compound mixed at one time using base resin and hardener of one lot number from one manufacturer to mold housings of plugs in one continuous, unchanged process. When a continuous type mixing process is used, a batch is that quantity of molding compound mixed and dispensed in one continuous unchanged process during a one (1) hour period or portion thereof using base resin and hardener of one lot number from one manufacturer.

6.5 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARDEC
Picatinny Arsenal, New Jersey 07806-5000
ATTN: AMSMC-QAT-M (D)

6.6 Subject (key word) listing.
Mine
Anti-personnel
Molding compound
Sensors
Mine functions
Breakwire
Anti-disturbance
Self-destruct
Integrated
Circuit timers

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project: 1345-A511)

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ELECTRONIC ASSEMBLY (P/N 93666912), TEST PROCEDURE

10 SCOPE.

10.1 Scope. This appendix details the text procedure to be used to electrically test the complete PDM Molded Assembly prior to its shipment. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20 APPLICABLE DOCUMENTS.

Government Drawings

9366690	System Schematic
9366692	Electronic Assembly

30 TEST REQUIREMENTS. All PDM molded assemblies containing electronic assembly 9366692 shall be tested in accordance with this document prior to shipment or performance evaluation. Only molded assemblies successfully passing this test shall be used for delivery against contract.

30.1 Absolute limits.

	VOLTAGE	CURRENT (RUN)	TEMPERATURE
MAXIMUM	6.0V	3.16 Milliamps	+125° ± 5°F
MINIMUM	--	--	-65° ± 5°F

30.2 Operating limits (hot and cold).

	VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC. FREQUENCY
MAXIMUM	4.5V	2.4 Milliamps	+60°C ± 5°C	Nominal* +5%
MINIMUM	2.7V	--	-25°C ± 5°C	Nominal* -5%

30.3 Operating limits (ambient).

	VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC. FREQUENCY
MAXIMUM	40V	2.2 Milliamps	+25°C	Nominal* +1%

*Nominal Frequency defined on Drawing 9298452.

30.4 V_B rise time and tolerance. The rise time on V_B shall be greater than one (1) millisecond and less than five-hundred (500) milliseconds to 96% of its final value. Tolerance on V_B is ± .05 volts.

40 SYSTEM TEST PROCEDURE. System tests shall be performed in accordance with Table I in a manner which meets the intent of the referenced requirement paragraph.

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30.1 Absolute limits.

	VOLTAGE	CURRENT (RUN)	TEMPERATURE
MAXIMUM	6.0V	3.16 Milliamps	+125° ± 5°F
MINIMUM	--	--	-65° ± 5°F

30.2 Operating limits (hot and cold).

	VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC. FREQUENCY
MAXIMUM	4.5V	2.4 Milliamps	+60°C ± 5°C	Nominal* +5%
MINIMUM	2.7V	--	-25°C ± 5°C	Nominal* -5%

30.3 Operating limits (ambient).

	VOLTAGE	CURRENT (RUN)	TEMPERATURE	OSC. FREQUENCY
MAXIMUM	40V	2.2 Milliamps	+25°C	Nominal* +1%

*Nominal Frequency defined on Drawing 9298452.

30.4 V_B rise time and tolerance. The rise time on V_B shall be greater than one (1) millisecond and less than five-hundred (500) milliseconds to 96% of its final value. Tolerance on V_B is ± .05 volts.

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40.1 Table I.

40.2 Test procedure.

40.2.1 Oscillator frequency. The oscillator frequency shall be measured by connecting the S.U.T. (System Under Test) as shown in Figure 1. The counter shall have an input impendence of 10 8 ohms or greater (C=10pF maximum) and the measurement shall represent a ten (10) period average. Frequency of both F1/4 and Fx shall be measured, and be within the limits of Table 1. Input F1/4 shall have a minimum of fifteen-hundred (1,500) fast counts applied and shall stabilize for five-hundred (500) milliseconds minimum before the measurement is made. Test sequence: 1) Close SW1/4 and allow fifteen-hundred (1,500) fast counts to enter; 2) Open SW1 and wait five-hundred (500) milliseconds minimum; 3) Close SW2 to F1; 4) Measure 10 period average of F1; 5) Close SW2 to F2; 6) Measure ten (10) period average of F2.

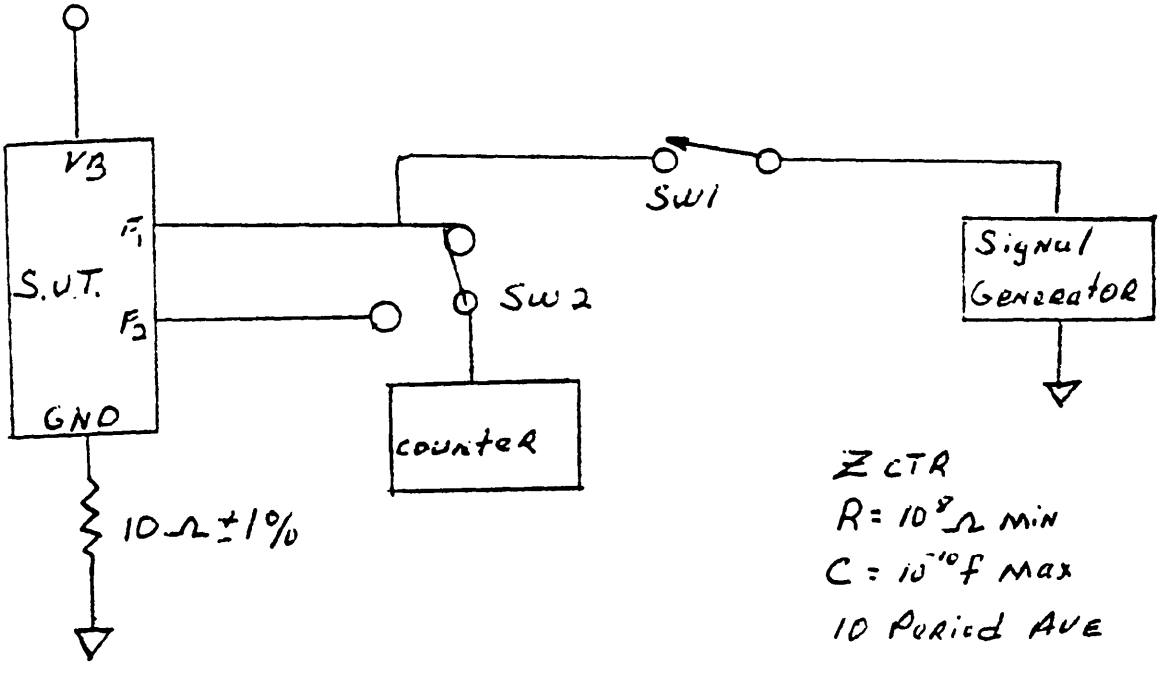


Figure 1. Oscillator Frequency Measurement.

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Table I

GROUP I (TA=+25°C) QUALIFICATION AND LOT ACCEPTANCE TESTS

TEST	TEST CONDITION	REQUIREMENT PARAGRAPH	LIMITS MAX.	MIN	UNIT
Output Capacitor Voltage*					
After 1 sec charge	VB=3.3V	40.2.6A	0.5	0.05	Volt
After 14 sec charge	VB=3.3V	40.2.6B		3.0	Volt
Detonator Waveform	VB/4=3.3V	40.2.4		Figure 6	
Initiator Waveform	VB/4=3.3V	40.2.4		Figure 6	
Oscillator Frequency	VB=4.0V	40.2.1	+2%	-1%	**
System Operating Current	VB=4.0V	40.2.2	2.40	1.8	Millam
Self-destruct Time Out	VB=3.3V	40.2.3	+1	-1	Count
L.V.D. Level		40.2.5	3.2	2.8	Volts (Type
Detonator Firing Circuit	VB=3.3V	40.2.7	2.00		Volt
Gas Generator Firing Circuit (Peak)	VB=3.3	40.2.8	2.00 1.00		lt Volts (Saturatio
Detonator Resistance		40.2.9	8.5	2.9	Ohms
Gas Generator/Piston Actuator Resistance		40.2.10	2.9	1.1	Ohms
Detonator Short		40.2.11	0.2		Ohms
Breakwire Continuity		40.2.12			
Loop 1			High Logic Level		
Loop 2			High Logic Level		
Capacitor Clamp	VB=3.3V	40.2.13	1.3	0.0	Volts 37 se
				2.5	Volts 50 se
Anti-disturbance Switch Operation		40.2.15	2.0		Vol

* With capacitor clamp circuit disabled by Pulling Test Point "C" to GND.
** Period value on Drawing 9298452.

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Table I

GROUP II (TEMPERATURE-VOLTAGE LIMITS)
QUALIFICATION TESTS ONLY

TEST	TEST CONDITION	REQUIREMENT PARAGRAPH	LIMITS MAX.	LIMITS MIN	UNITS
Tests to be run at T_A= -25°C V_B=4.5V, 3.5V and T_A= +60°C V_B=4.5V, 3.3V unless otherwise specified					
Detonator Waveform		40.2.4	Figure 6		
Gas Generator Waveform		40.2.4	Figure 6		
Oscillator Frequency	V_B=4.2 Upper Limit V_B=3.5 Lower Limit	40.2.1	+5%	-5%	o
System Operating Current		40.2.2	2.4	1.8	Milliamps
Self-destruct Time Out		40.2.3	+1	-1	Count
L.V.D. Level		40.2.5	3.6	2.4	Volts
Detonator Firing Circuit		40.2.7		1.6 Volts (Peak) .8 Volts (1MS)	
Gas Generator Firing Circuit		40.2.8		1.6 Volts (Peak) .8 Volts (1MS)	

o Period value on Drawing 9298452.

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40.2.2 System operating current. The system operating current shall b measured by connecting the S.U.T as shown in Figure 2. Input F_1 shall have fifteen-hundred (1,500) fast counts applied with SW1 closed. Open SW_1 one-hundred (100) milliseconds minimum and measure operative current. Operating current must meet the requirements of Table II.

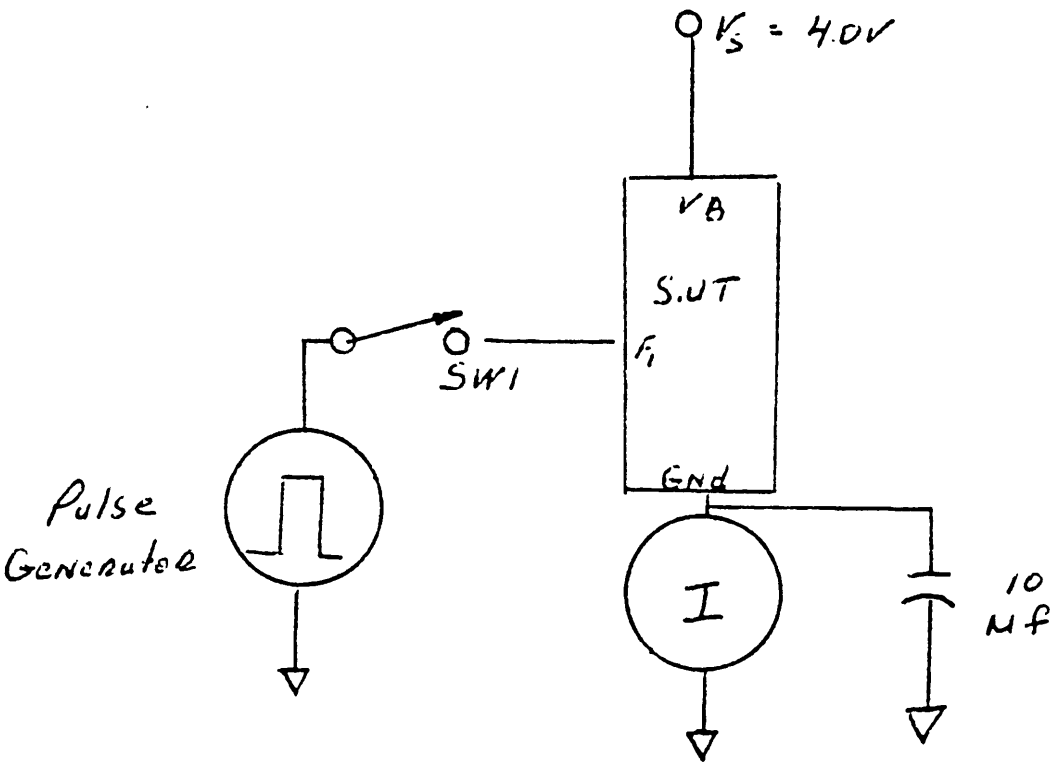


Figure 2. Operating current measurement.

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40.2.3 Self-destruct time out. The self-destruct time out shall be tested by connecting the S.U.T. as shown in Figure 3. Input F_1 shall have pulses applied at a rate consistent with the overall testing time desired and the output shall be observed at F.

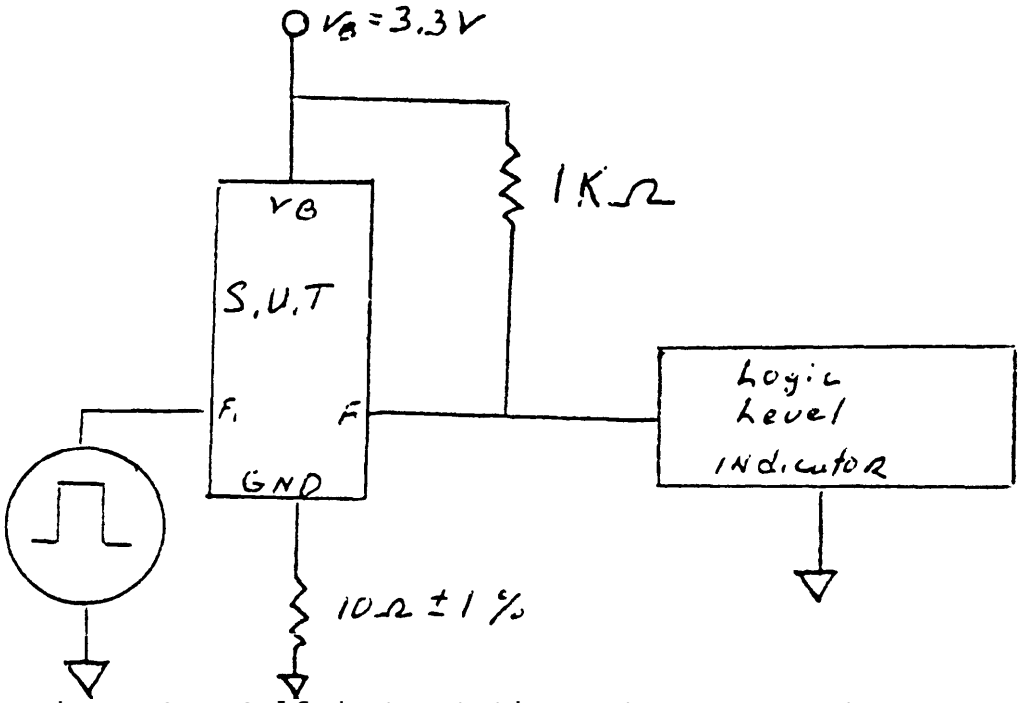


Figure 3. Self-destruct time out measurement.

TEST SEQUENCE:

1. Connect S.U.T. as shown in Figure 3.
2. V_B turn on time shall be as specified in paragraph 3.4.
3. PUlse generator shall start when V_B 3.3V at a convenient repetition rate. The pulse generator shall have a duty cycle of 30% to 70%.
4. Output shall switch from a logic high (V_B) to a logic low (GND) after 32,000 input counts:

NOTE: If time out is measured by time rather than counts, then the limit shall be ±0.1%.

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40.2.4 Detonator and gas generator logic. The S.U.T. shall be connected as shown in Figure 4. When the input waveforms shown in Figure 5 are applied, the output shown in Figure 6 shall occur.

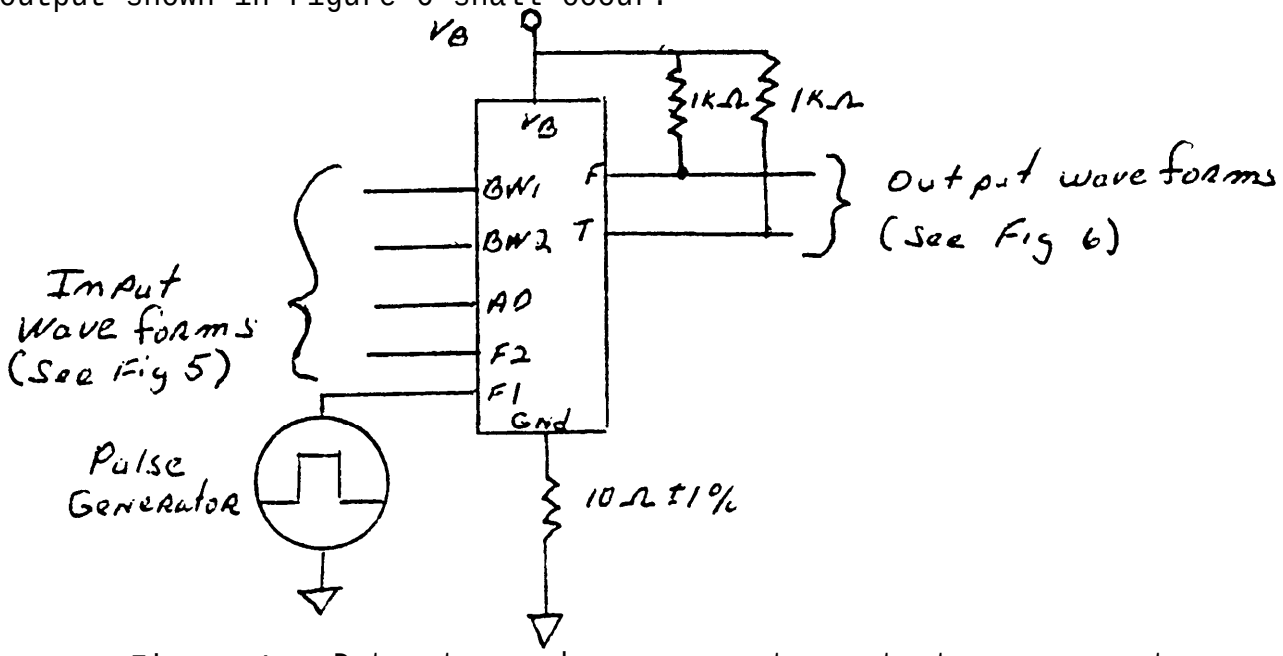


Figure 4. Detonator and gas generator output measurement.

The test sequence shown in Figure 5 and 6 may be changed provided all tests specified are accomplished. The order of accomplishment is not important. The "T" output waveform shall be present when F_1 = One-hundred and sixty (160) pulses. However, only one (1) pulse shall be produced at "T"

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Input wave forms

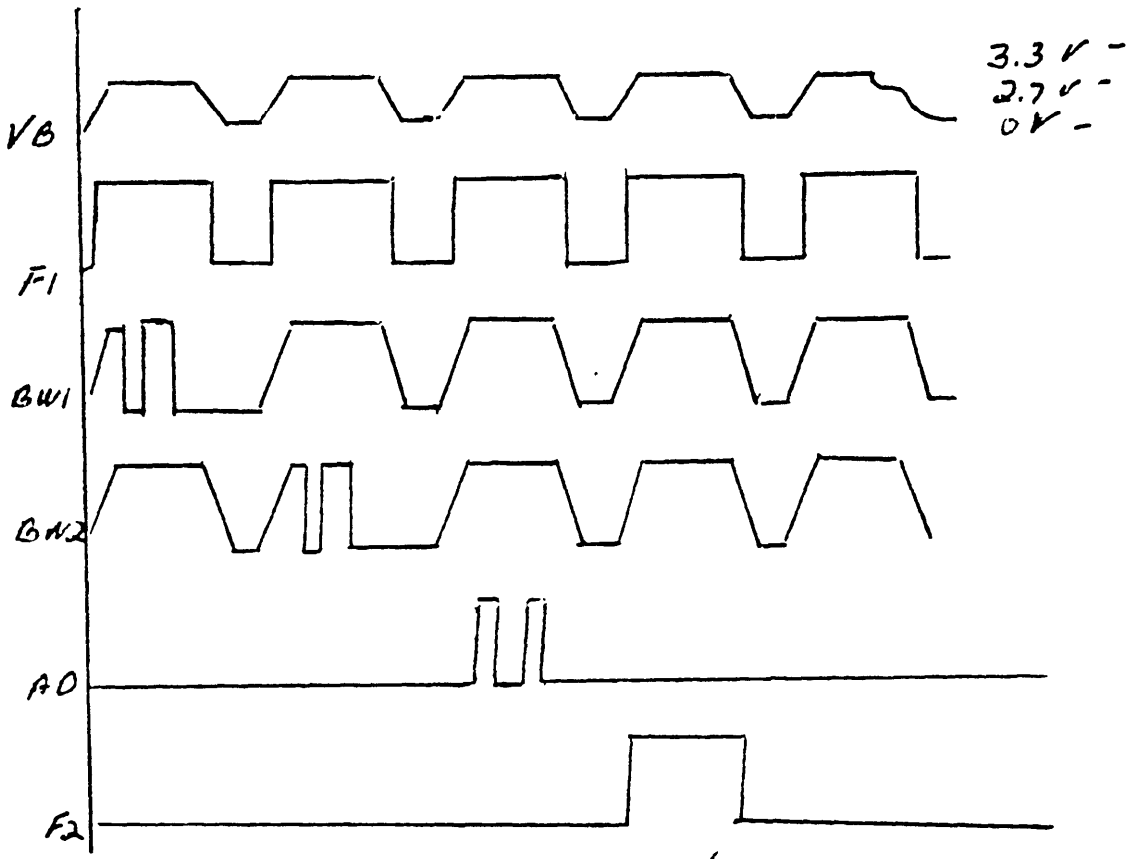


Figure 5 Input Wave Forms

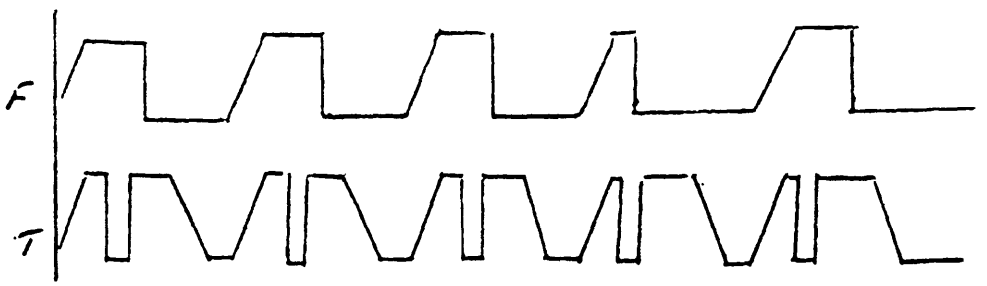


Figure 6 Output wave forms

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40.2.5 L.V.D. level. With the S.U.T. connected as shown in Figure an V_B=4.0V, a minimum of one-hundred and seventy-six (176) pulses shall be applied on the F_1 input. The voltage on V_B shall be decreased at 100mv/SEC maximum until an output occurs on the F output. The voltage (VB) a which this occurs shall be recorded as the L.V.D. voltage.

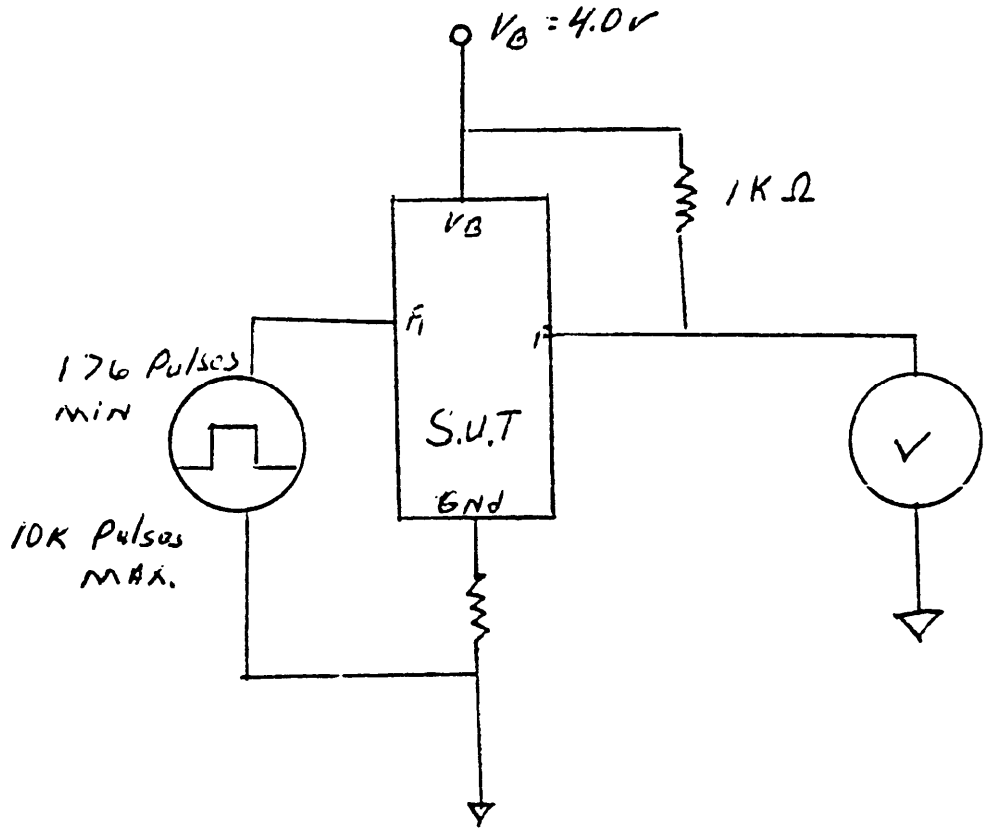


Figure 7. L.V.D. level measurement.

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40.2.6 Output capacitor voltage. The voltage on the output capacitor shall be measured by connecting the S.U.T. as shown in Figure 8.

a. Capacitor present - with SW_1 open, apply power and verify that V_C rises to the limits specified in Table I. Take this measurement one (1) second after power on.

b. Capacitor leakage - Close SW1 until V_C=3.3 volts, then open SW_1. Upon opening SW_1, wait fourteen (14) seconds minimum then measure V_C V_C must be greater than the value specified in Table I.

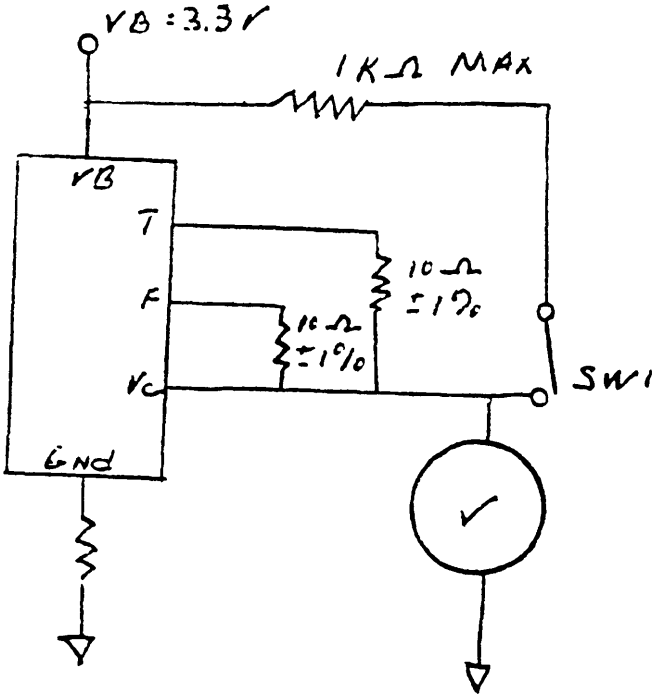


Figure 8. Output capacitor voltage measurement.

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40.2.7 Detonator firing circuit. The S.U.T. shall be connected as in Figure 9.* A minimum of 176 pulses shall be input on F_1 followed by 1 pulse input on Z. The output on F shall be measured and shall meet the requirements in Table I, after Z has an input.

* Quick charge the capacitor to V_B by closing SW_1 before the input of 176 pulses. The open SW_1 before giving an input to Z and measuring the output on F.

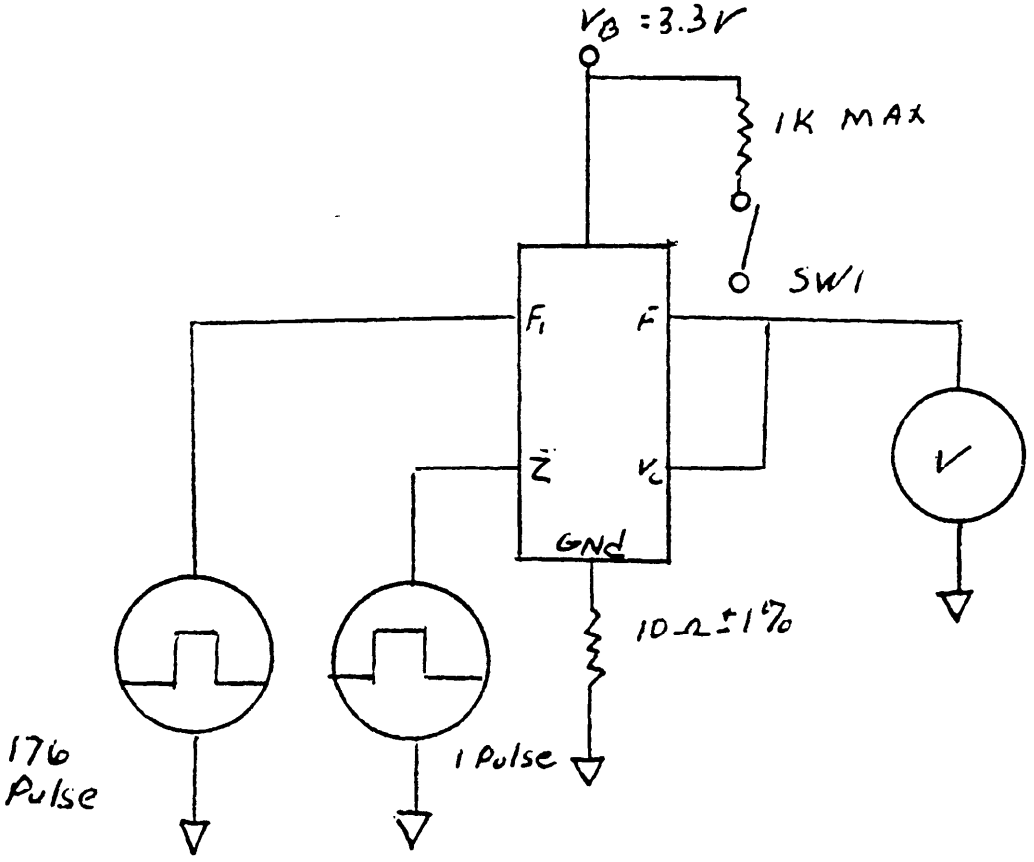


Figure 9. Detonator output measurement.

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40.2.8 Gas generator firing circuit. The S.U.T. shall be connected as shown in Figure 10.* A total of 160 pulses shall be input on F_1. The output on T shall be measured and shall meet the requirements in Table I.

*Quick charge the output capacitor by closing SW_1 before input of 160 pulses. Open SW_1 before applying 160 pulses in F1.

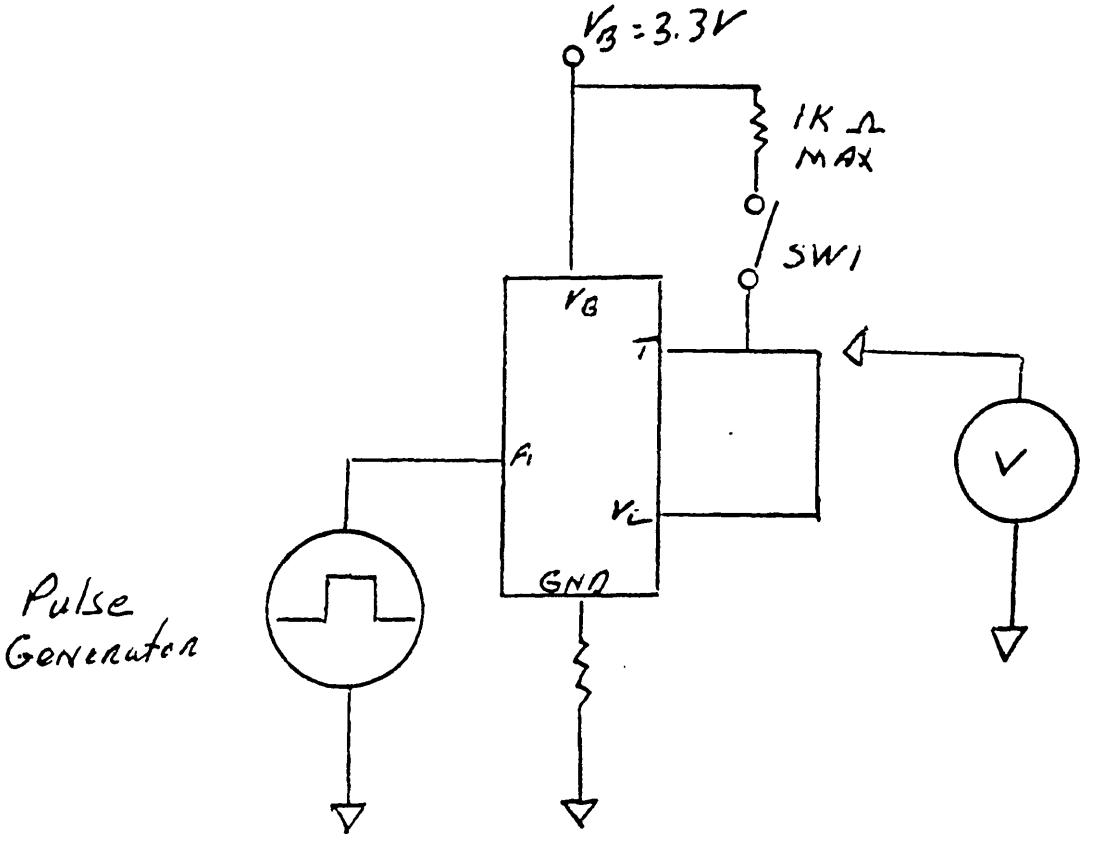


Figure 10. Gas generator output measurement.

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40.2.9 Detonator resistance. The detonator resistance shall be measured using an ohmmeter whose current output if 5.0 ma maximum between F and S. The resistance measured shall be between the limits specified in Table I.

40.2.10 Gas generator//Piston actuator resistance. The initiator in parellel with the piston actuator resistance shall be measured using an ohmmeter whose current output is 5.0 ma maximum between T and V_C (or preform). The resisitance measured shall be between the limits specified in Table I.

40.2.11 Detonator short. The detonator short shall be measured using an ohmmeter whose current output is 5.0 ma maximum between F and V_C (or preform). The resistance shall be within the limits specified in Table I.

40.2.12 Breakwire continuity. The breakwire continuity shall be measured by connecting the S.U.T. as shown in Figure 11. A high logical output must be present at both BW_1 and BW_2.

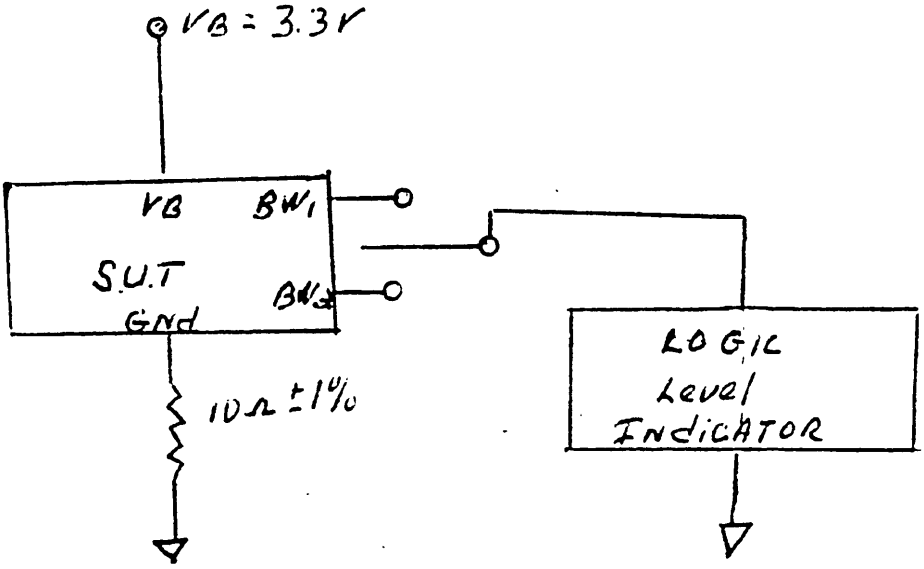
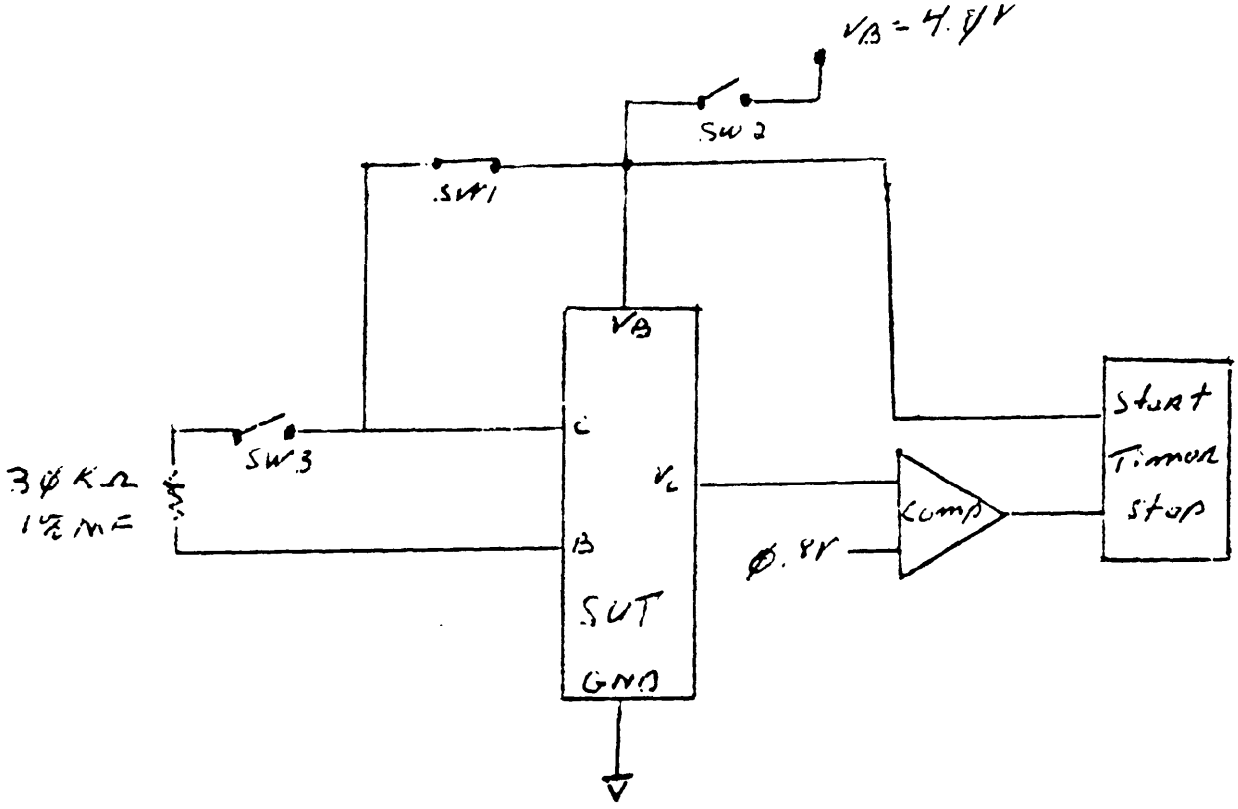


Figure 11. Breakwire continuity check.

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40.2.13 Capacitor clamp time delay measurement. The capacitor clamp time delay shall be measured by connecting the S.U.T. as shown in Figure 12. Before closing SW_2, SW_1 shall be clocked for 20 seconds minimum. SW_1 will then be opened and SW_2 closed.

When SW_2 is closed, the timer will start counting, and will continue to count, until V_C rises above 0.8 volts at which time the voltage comparator will stop the timer. The time delay measured must meet the requirements for Normal Time Delay in Table II. To measure Short Time Delay, close SW3 before starting test.



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40.2.14 Anti-disturbance switch operation. With the S.U.T. connected as shown in Figure 13, the S.U.T. shall be moved to verify anti-disturbance switch function.

After the input of 180 pulses, the S.U.T. shall be moved through an arc of 30° minimum. After movement is complete, an output logic "0" shall be measured.

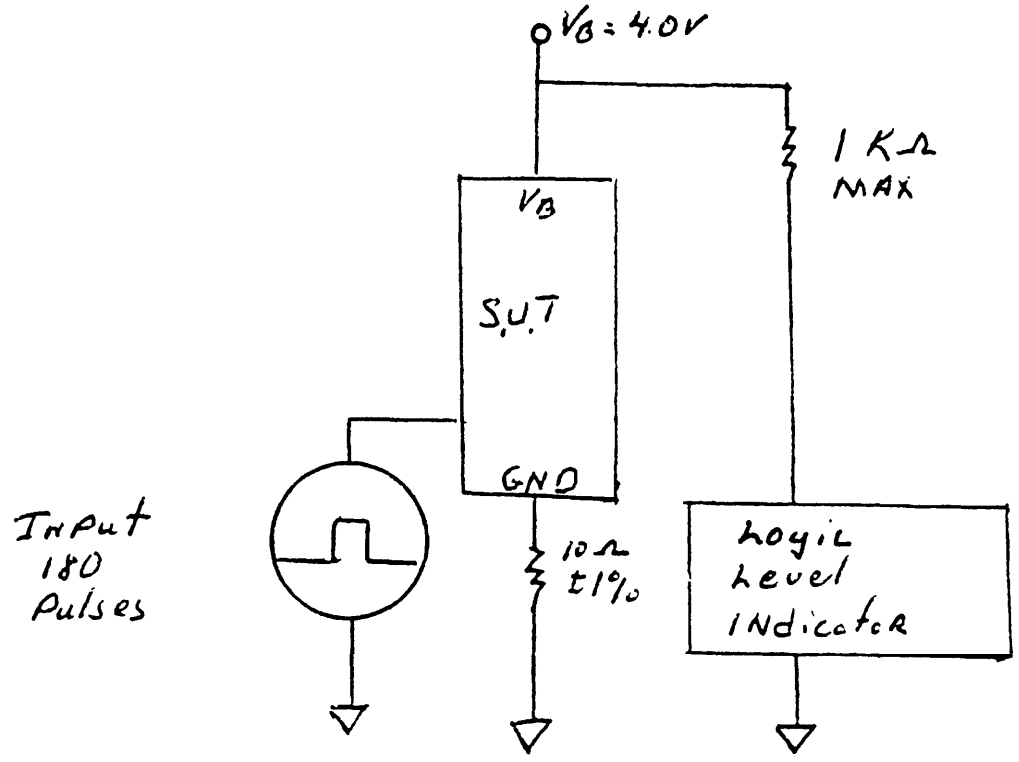


Figure 13. Anti-disturbance switch output measurement.

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